

LPS / PS-LPS Program: Plan of Action and Phase Sequencing

A dependency-ordered roadmap across validation, synchronization, local-dimension regularization, local covariance and the simplicial-complex substrate, synthetic 16S data, and the occupancy / quasi-equilibrium decomposition

Internal program plan — LPS / `geosmooth` program

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Abstract

The audit of the LPS/PS-LPS effort opened into a layered research program. This note fixes the *order* in which its pieces should be built, with the dependencies made explicit so the sequence is a consequence of what each piece needs rather than a preference. The spine follows the components as listed — validation first, then the estimator extensions, then the compositional/covariance layer and its simplicial-complex substrate, and finally the occupancy / quasi-equilibrium decomposition — with three structural refinements: (i) the synthetic-data and VALENCIA-exploration workstream is pulled *forward* as a cross-cutting enabler, because it is a dependency of almost every later test, not a last-bundle item; (ii) PS-LPS synchronization and local-dimension regularization run *in parallel* as independent extensions of the validated base, then combine; and (iii) the simplicial complex (the nerve of the chart cover) is named as the *integration* phase that unifies synchronization, dimension, and covariance, not as a sub-part of any one of them. Each phase is specified by its objective, inputs (dependencies), key deliverables (the existing design notes are its specs), and a go/no-go gate; the whole program runs under the two-agent audit-independence workflow. The immediate next actions are the two audits already queued: the LPS Tier-0 run and the LPS binary GM/FF run.

On the figures. Both figures are original schematics drawn for this note.

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1 How to read this, and the one rule that wraps every phase

This is a sequencing document, not a methods document: the methods live in the design notes referenced at the end, and each phase below points to the note that specifies it. The job here is only to say what is done first, second, and so on, and why the order is forced.

One rule wraps every phase. Each phase is executed under the *two-agent audit-independence workflow*: an implementer builds, an independent auditor validates against a fixed charter (the auditor’s scope is not set by the implementer, the handoff carries evidence and admitted limitations but not the audit’s questions, and “report rendered” is never “phase accepted”). A phase is not complete — and the next phase does not start — until its go/no-go gate is passed by that independent audit or adjudicated by the investigator. This is what keeps a fast-moving, multi-layer program from accumulating unvalidated machinery.

Why this ordering The ordering is driven by *dependencies*, not by importance or interest. A phase may begin only once every input it draws an arrow from in Figure 1 has passed its gate. Two phases with no dependency between them run in parallel. The capstone is last because it depends on the most other pieces, not because it is an afterthought.

2 The components, and how the dependencies fall

The program has five technical components plus one cross-cutting enabler and one terminal goal. Stated as the dependency structure of Figure 1:

- **Phase 0 — LPS validation.** The base smoother, proven correct (polynomial reproduction, the linear-smoother identity, df, honest cross-validation, boundary behaviour, binary calibration, degenerate-geometry robustness). It *emits* a toolkit — the smoother matrix S , $\text{tr } S$, analytic leave-one-out / GCV, and uncertainty bands — that later phases reuse. Foundation for everything.
- **Cross-cutting — the data enabler.** Synthetic non-manifold generators with known ground-truth dimension/components, maturing from toy stratified objects to VALENCIA-realistic 16S generators via the E1/E2/E3 exploration. A *dependency* of the dimension field, the covariance layer, and the occupancy capstone — hence front-loaded, not last.
- **Phase 1a — PS-LPS synchronization.** Prediction-overlap coupling of local models. An independent extension of the validated base; reuses the Phase-0 toolkit for the df-matched comparison and exact λ_{sync} selection.

- **Phase 1b — local-dimension regularization.** The ℓ_1 /TV dimension field with exact graph-cut solve. Independent of 1a; needs the validated base and the synthetic generators.
- **Phase 2a — local covariance / association.** The chart-aware covariance/precision layer on the compositional (L^∞) geometry; needs the base and the realistic 16S generators.
- **Phase 2b — the simplicial complex (nerve): integration.** The nerve of the chart cover, unifying $1a + 1b + 2a$ into one consistency objective over the overlaps; depends on all three.
- **Phase 3 (capstone) — occupancy / quasi-equilibrium / support decomposition.** Finding pure components and transition phases; builds on the covariance and the nerve and the realistic generators.
- **Terminal — the real 16S / omics application.** The actual estimation on real data, which the whole program exists to serve.

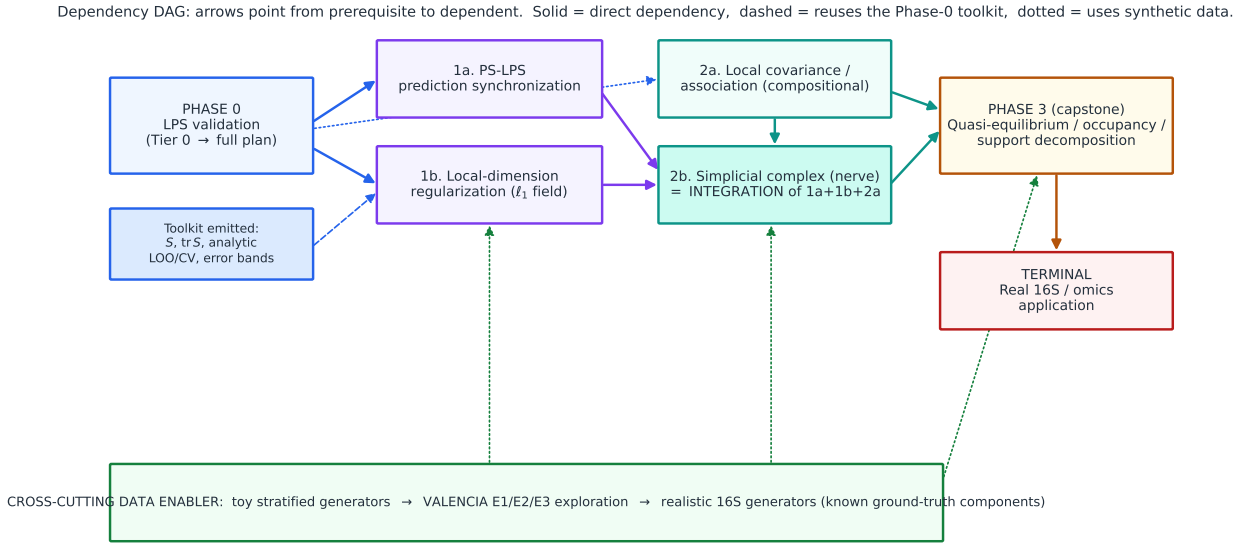


Figure 1: The program dependency DAG. Arrows point from prerequisite to dependent: solid = direct dependency, dashed = reuses the Phase-0 toolkit, dotted = uses the synthetic data. PS-LPS (1a) and the dimension field (1b) are parallel; the nerve (2b) integrates $1a+1b+2a$; the occupancy capstone (Phase 3) and the terminal application sit downstream of the integration.

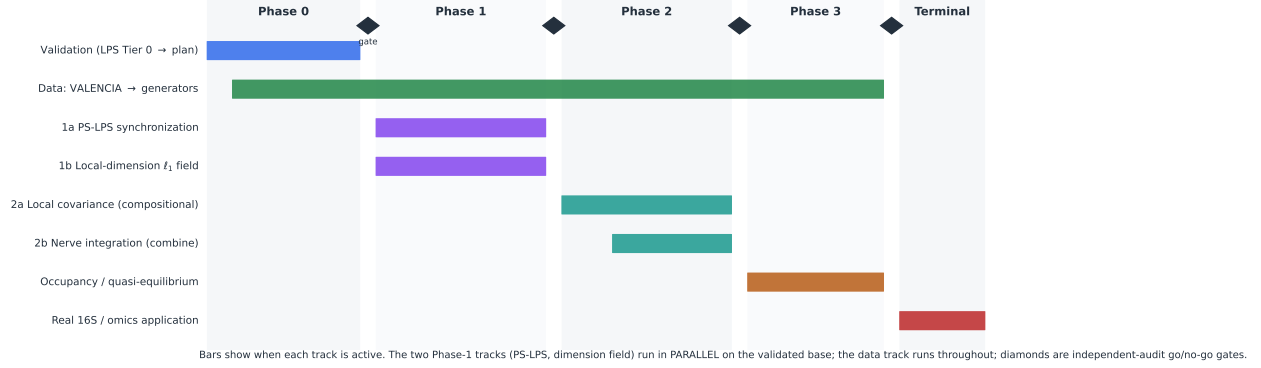


Figure 2: The same plan as parallel tracks over time. Validation runs first; the data enabler runs throughout; the two Phase-1 extensions run in parallel on the validated base; the covariance and nerve-integration tracks follow; the occupancy capstone and then the real-data application close the program. Diamonds are independent-audit go/no-go gates between phases.

3 Phase 0 — LPS validation (the foundation that emits the tools)

Objective. Prove the base smoother is what it claims, and produce the reusable estimator toolkit.

Inputs. None (the foundation). **Deliverables.** The Tier-0 correctness battery (polynomial reproduction; the linear-smoother identity $\hat{y} = Sy$ with $\text{tr } S$ and analytic LOO/GCV; boundary behaviour; consistency; binary recovery and calibration; CV no-leakage; degenerate geometry), then the rest of the experimental plan (honest nested/grouped CV; bandwidth/kernel semantics; binary-mode hygiene; uncertainty bands). Spec: the *experimental plan* note. **Gate.**

Go/no-go gate The Tier-0 battery passes deterministically; the linear-smoother toolkit (S , df, LOO/GCV, bands) is implemented and tested. No later phase begins until this gate is green — it is both the correctness gate and the source of the tools the df-matched, GCV, and band computations downstream require.

Unblocks. Everything — and specifically hands PS-LPS its df-matched comparison and exact λ_{sync} selection.

Immediate next action. The two audits already queued are the first concrete execution of this phase: (1) the **LPS Tier-0 run** (does the correctness battery actually pass, and is the toolkit correct?), and (2) the **LPS binary GM/FF run** (the binary-mode behaviour, with the fallback-telemetry and selection-metric fixes from the earlier results audit). Both are Phase-0 evidence; neither downstream phase should start before they are adjudicated.

4 Cross-cutting — the synthetic-data and VALENCIA-exploration enabler

Objective. Supply known-ground-truth, increasingly realistic non-manifold datasets, because every downstream test needs an answer key the real data cannot give. **Inputs.** Phase 0 (charts), and — for the realistic versions — the VALENCIA relative-abundance matrix.

Deliverables. Early: the toy stratified generators (the implementation notes' DGP library) needed by Phase 1b. Maturing: the VALENCIA E1 (enumerate naive strata), E2 (mass concentration near lower faces), E3 (parameter

bundle) exploration, producing the realistic stratum-mixture generators with known components that feed Phases 2 and 3. Spec: the *implementation notes*. **Gate.**

Go/no-go gate Toy generators ready before Phase 1b’s boundary/scale tests. Realistic generators pass a posterior-predictive fidelity check (synthetic marginals reproduce the real VALENCIA marginals) before any Phase-2/3 test built on them is trusted.

Unblocks. Phase 1b (toy), Phase 2a and Phase 3 (realistic).

Why this ordering This is the one item promoted out of the original ordering. In the initial list, synthetic models sat bundled with the covariance layer near the end. But the dimension-field tests (1b), the covariance tests (2a), and the occupancy capstone (3) all consume known-truth synthetic data, and the realistic versions need the VALENCIA exploration to exist first. A dependency cannot trail the things that depend on it, so the data workstream leads and runs throughout.

5 Phase 1 — two parallel extensions of the validated base

PS-LPS (1a) and the local-dimension field (1b) are *independent* of each other: neither uses the other’s output. Both depend only on Phase 0 and the synthetic generators. They therefore run in parallel and are combined later, in Phase 2b — which is exactly the “test independently, then combine” structure the program already anticipated.

5.1 Phase 1a — PS-LPS prediction synchronization

Objective. Validate, and put on firm footing, the synchronization of overlapping local models through prediction-overlap regularization. **Inputs.** Phase 0 (the validated base *and* the toolkit).

Deliverables. Exploit the fact that PS-LPS is linear in the response: implement exact effective $df = \text{tr } S(\lambda_{\text{sync}})$ and analytic LOO/GCV in place of the K -fold materialised CV, and compare PS-LPS to LPS at *matched df* (not matched λ); fix the ridge-path discontinuity at $\lambda_{\text{sync}} = 0$; make the synchronization graph (currently a fixed kNN-rank threshold) and its weights explicit; characterize the estimand and the $\lambda_{\text{sync}} \rightarrow \infty$ limit; add uncertainty bands. Spec: the PS-LPS sections of the audit memo and the chart-aware synchronization. **Gate.**

Go/no-go gate On synthetic data, PS-LPS improves over a *df-matched* LPS where the geometry predicts it should; λ_{sync} selection is well-founded via $\text{tr } S$ (no longer railing to a grid boundary for lack of a df-aware criterion); the $\lambda_{\text{sync}} = 0$ gate reproduces LPS exactly and the path is continuous.

Unblocks. Phase 2b (integration over the nerve).

5.2 Phase 1b — local-dimension regularization

Objective. Validate the ℓ_1 /total-variation local-dimension field and its exact graph-cut solver.

Inputs. Phase 0; the toy stratified generators. **Deliverables.** The test battery T1–T9: graph-cut exactness, ℓ_1 -keeps- the-boundary vs ℓ_2 -blurs, denoising, the boundary safeguard, scale/persistence,

the ξ -vs- u coordinate trap, the support cap and soft face, the frontier (no impossible islands), and downstream Truth-RMSE. Spec: the implementation notes. **Gate.**

Go/no-go gate The implementation notes’ single go/no-go: *(the E2 nominal–effective gap is large) $\wedge$ (T9 improves Truth-RMSE on the stratum-mixture generator)*. If either fails, the within-face dimension field is not carried to production; the cheaper soft-face-and-cap refinement is kept instead.

Unblocks. Phase 2b.

6 Phase 2 — the compositional covariance, and its integration

6.1 Phase 2a — local covariance / association on the simplex

Objective. Build and validate the local covariance/precision (the “shape” of the local structure) on the boundary-safe compositional geometry. **Inputs.** Phase 0; the realistic 16S generators; a compositional front end (a boundary-aware log-ratio / Aitchison chart with structural-zero and soft-face handling — a prerequisite this phase must establish first). **Deliverables.** Recovery of a known local covariance on synthetic 16S; the two-layer association object (co-occurrence support layer + conditional-abundance precision layer) with the soft-face and ℓ_1 -fusion refinements. Spec: the chart-aware association design and its addendum. **Gate.**

Go/no-go gate The covariance/precision is recovered on synthetic data with known structure; the boundary-safe path never reintroduces a pseudocount; the ℓ_1 fusion preserves a planted reorganization across a community-state boundary that ℓ_2 would smooth away.

Unblocks. Phase 2b and the capstone.

6.2 Phase 2b — the simplicial complex (nerve): integration

Objective. Unify the three preceding layers — prediction synchronization (1a), the dimension/size field (1b), and the covariance (2a) — on the nerve of the chart cover, as one consistency objective over the overlaps. **Inputs.** Phases 1a, 1b, 2a. **Deliverables.** The nerve-substrate program, minimal-first: replace PS-LPS’s pairwise overlap graph with the nerve 1-skeleton of variable-size charts; add 2-simplices (higher-order synchronization at junctions); co-regularize chart size and dimension over the nerve; and, if it pays, the single cochain objective combining prediction, covariance, and dimension. Spec: the chart-nerve substrate note. **Gate.**

Go/no-go gate On synthetic data, the nerve 1-skeleton of variable-size charts beats the fixed kNN overlap graph on held-out accuracy; higher-order (2-simplex) synchronization either measurably helps (at junctions) or is dropped; the integrated objective is validated on known truth.

Unblocks. Phase 3 — it supplies the basin-and-corridor skeleton the capstone runs on.

7 Phase 3 (capstone) — occupancy / quasi-equilibrium / support

Objective. Find the pure components (recurrent ecological basins) and the transition phases between them. **Inputs.** Phase 2a (covariance), Phase 2b (the nerve / basin-corridor skeleton), the realistic 16S generators, and the COT occupation-measure machinery. **Deliverables.** The layered route: level-set basins on the chart cover (an identifiable component finder), the geometric corridor discriminator (overlap mass + elongated covariance), the local-covariance basin shape, the alignment of per-subject basins onto shared components, and the transfer operator on the coarse basin-and-corridor skeleton that supplies the dynamics and earns the “quasi-equilibrium” label. Spec: the COT bridge note. **Gate.**

Go/no-go gate On synthetic 16S with *known* components: the geometry-first level-set basins are more stable across scales and bootstraps than the dictionary-factorization atoms; the transfer-operator metastable count agrees with the cluster-tree count; the corridor discriminator removes spurious components. Then, and only then, the HMP/152-subject experiment.

Unblocks. The terminal application.

8 Terminal — the real 16S / omics application

Objective. Do the thing the program exists for: estimate conditional expectations of binary and continuous features, and the community occupancy/component structure, on real 16S and other omics data, with honest uncertainty. **Inputs.** All prior phases. **Deliverables.** Real-data conditional-expectation surfaces with bands; occupancy/component decompositions; reported under nested and grouped (subject-level) cross-validation so the headline numbers are not selection-optimistic or leakage-inflated. **Gate.**

Go/no-go gate Held-out predictive validity on real data and biological interpretability of the components — judged, as throughout, by prediction, not by intrinsic-geometry scores.

9 Where this diverges from the initial ordering, and why

The initial list was: (1) LPS validation, (2) local-dimension regularization, (3) PS-LPS, (4) covariance + simplicial complex + synthetic data, (5) occupancy. The spine of that ordering is right and is preserved. The refinements are:

1. **Validation first — agreed, emphatically**, and for a stronger reason than “it gates the rest”: Phase 0 *emits* the $S/\text{df}/\text{LOO}/\text{bands}$ toolkit that the df-matched PS-LPS comparison, the dimension field’s GCV, and the covariance bands all reuse. It is the foundation that hands the later phases their tools.
2. **Data enabler pulled forward** (§4). It was bundled near the end of the list; it is in fact a dependency of items (2), (4), and (5), so it must lead.
3. **PS-LPS and the dimension field run in parallel**, not in sequence. They are independent extensions of the validated base; the list put one before the other, but nothing connects them until they are combined in Phase 2b.

4. **The simplicial complex is the integration phase (2b)**, not a sub-part of the covariance work. It is where synchronization, dimension, and covariance are unified over the nerve.
5. **The terminal real-data application is named explicitly** as the endpoint, and **the audit-independence workflow wraps every phase** as the governance layer.

10 Decision gates that can re-route the program

The plan is adaptive, not a fixed waterfall: several gates can prune whole branches.

- **After E2 (data).** If the nominal–effective dimension gap in the real VALENCIA strata is small, the within-face dimension field (1b) is deprioritized to the cap-and-soft-face refinement, and 2a/3 proceed without it.
- **After T9 (1b).** If the dimension field does not improve Truth-RMSE, it is not promoted to production; a global chart dimension is kept.
- **After 2b.** If higher-order (2-simplex) synchronization does not measurably help, the program stays on the nerve 1-skeleton.
- **After Phase 3 on synthetic.** If the geometry-first basins are not more stable than the existing factorization, the capstone falls back to the COT program’s current dictionary route rather than replacing it.

11 Immediate next actions

1. **Audit the LPS Tier-0 run** — does the correctness battery pass, and is the linear-smoother toolkit correct? (Queued; you will share the results.)
2. **Audit the LPS binary GM/FF run** — the binary-mode behaviour, with the fallback-telemetry and selection-metric fixes from the earlier results audit.
3. **In parallel**, confirm the toy stratified generators are ready, so Phase 1b’s boundary and scale tests have their known-truth inputs the moment Phase 0 clears its gate.

Once these two audits are adjudicated, Phase 0 is gated and the parallel Phase-1 tracks (PS-LPS and the dimension field) begin together.

References

- [1] *LPS Correctness and Validation: A Precise Experimental Plan* (Phase 0 spec), `~/current_projects/geosmooth/project_briefs/lps_experimental_plan_2026-06-09`.
- [2] *LPS/PS-LPS JASA-style audit memo* (PS-LPS linearity, df, λ_{sync} selection; the binary-run results audit), `~/current_projects/geosmooth/project_briefs/lps_ps_lps_jasa_style_audit_2026-06-09`.
- [3] *LPS Local Dimension Regularization: Implementation Notes* (Phase 1b spec; the data enabler E1/E2/E3 and synthetic generators), `~/current_projects/geosmooth/project_briefs/lps_local_dimension_regularization_implementation_notes_2026-06-10`.
- [4] *Chart-Aware Local Association Structure on the L^∞ Atlas* (Phase 2a spec), `~/current_projects/geosmooth/project_briefs/chart_aware_local_association_design_2026-06-09`.
- [5] *The Local-Chart Simplicial Complex as an Organizing Substrate for LPS* (Phase 2b spec), `~/current_projects/geosmooth/project_briefs/lps_chart_nerve_substrate_2026-06-10`.

- [6] *Occupancy Supports as Basins on the Chart Cover* (Phase 3 spec), `~/current_projects/vaginal_community_trajectory_types/docs/occupancy_basins_on_chart_cover_bridge_2026-06-10`.
- [7] *Two-Agent Research Project Workflow* (the governance wrapper / audit-independence rules), `~/codex/notes/workflows/two_agent_research_project_workflow`.